Figure 3: Kubeflow platform

Personalised content recommendation: Media and entertainment platforms utilise ML models to recommend personalised content to users. MLOps enables the seamless delivery of content recommendations in real-time, enhancing user engagement and satisfaction.

In all these cases, MLOps plays a crucial role in automating the ML model life cycle, including model training, testing, deployment, monitoring, and updates. This streamlines the development process, reduces manual errors, enhances collaboration between teams, and allows organisations to deliver high-quality ML-powered applications more efficiently and reliably.

Prominent platforms for MLOps

Many platforms offer a robust ecosystem to streamline machine learning workflows. Two major ones are Kubeflow and TensorFlow Extended.

Kubeflow

URL: https://www.kubeflow.org/

This is an open source platform built on Kubernetes, designed to streamline the deployment, scaling and management of machine learning workloads. It provides tools for building, training, and deploying models in a consistent and reproducible manner. Kubeflow leverages Kubernetes' container orchestration capabilities to enable seamless scalability of ML workloads. This means that as the demand for machine learning models grows, Kubeflow can efficiently allocate resources and scale the infrastructure to meet it. This elasticity ensures that ML models can handle various workloads, from small-scale experimentation to large-scale production deployments.

In MLOps, it is vital to have a well-documented and version controlled process for model training. Kubeflow provides tools and components that allow data scientists to package their ML code, dependencies, and configurations in a consistent manner using containers. This ensures that models can be easily reproduced and versioned, facilitating collaboration among teams and reducing the risk of model discrepancies in production.

TensorFlow Extended (TFX)

URL: https://www.tensorflow.org/tfx

Developed by Google, TFX is an end-to-end platform for deploying production machine learning pipelines. It includes components for data validation, preprocessing, model training, and serving. It plays a crucial role in the implementation of MLOps by enabling the end-to-end management of machine learning workflows. By seamlessly integrating TensorFlow, TFX facilitates the efficient development, deployment, and monitoring of machine learning models in real-world applications. At its core, TFX is designed to address the complexities associated with scaling and operationalising machine learning projects. It encompasses a collection of modular components, each serving a specific function within the pipeline. These components work cohesively to automate the machine learning life cycle, from data ingestion to model serving and beyond.

Other platforms for MLOps

There are other, lesser-known platforms too that play a vital role in shaping the MLOps landscape.

MLflow

URL: https://mlflow.org/

This open source platform facilitates the management of the complete machine learning life cycle. MLflow allows you to track experiments, package code into reproducible runs, and share and deploy models.

Continuted to page...35